Estimating Portugal's Wine Demand Using the Almost Ideal Demand System

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1 Introduction

The history of wine has been linked to the history of mankind for as long as humanity's collective memory can remember. Every major civilization has had a rich history with the magic elixir. The earliest confirmed remains of a winery come from before 4000BC. However, when modern readers hear words like wine and wineries, they almost immediately form pictures in the mind of the Mediterranean region. This region holds a rich history in wine production—and consumption—that is rivaled by none. The area of the western Mediterranean, in particular, composed of the countries of Italy, France, Spain, and Portugal are world renowned for their special blends. This region, from the Iberian Peninsula in the West to the Italian Peninsula to the East is the "Fertile Crescent" of wine production. Italy is known for its Barolo, Lambrusco, and many others. France for its famed Champagne region. Spain for its Rioja and Sherry and Portugal for Port.

This paper seeks to study this diverse market through the lens of a demand system in order to give insights into the industry for importers, exporters, and policymakers. This study also attempts to show the danger in considering a commodity such as wine as a homogeneous product. As the results show, this could not be farther from the truth, with wines from each region showing very different characteristics in consumer demand.

This paper focuses on the wine industry, in particular, because of the clear fact that country of origin plays a large role in all aspects of consumer preferences for wine. Preferences vary based on type of grape, winery location, and perceived quality. All of which can clearly be seen as dependent on country of origin. Take Champagne for instance. Two identical barrels could be fermented mere miles from each other—but if one is lucky enough to be within the bounds of the Champagne-Ardenne region of France, it is called "Champagne," while the other, not being so lucky, will be stuck with the name of "Brut." The different location of those two barrels completely separates the wine inside them into two categories. Whereas in other markets there can be many different brand loyalties based on subjective preferences, the fact of locational preference in the wine industry allows economists to form robust estimations using only country-wide data instead of needing more specific brand categories.

The choice to look at the country of Portugal was motivated by several factors. First, Portugal has a long and rich history in wine-making and consumption. Wine making was introduced to the Iberian peninsula nearly three thousand years ago by the Phoenicians (Estreicher, 2013). While wine is inseparable from the history of Portugal, what makes it really stand out is how that history plays out in the present day. In a 2020 study, Portugal rose to the top of the list in per capita wine consumption by country, its citizens on average consuming over 50 liters of wine each year.[1] Portugal is further an interesting case study because while it is clearly a country that has a taste for wine, it is not one of the big three producers (France, Italy, and Spain), which allows us to look at their imports from these countries. Finally, the fact that Portugal, France, Italy, and Spain are all in the European Union allows us to compare these countries equally without the worry of differing tariff levels (as the EU has free trade within its bounds). This allows for a clean apples-to-apples comparison of how the Portuguese people view the product of these three wine producing

[1] https://www.bkwine.com/features/more/wine-consumption-per-person-2020/

giants.

The empirical method I use for this study is the Almost Ideal Demand System (AIDS) model, first introduced by Deaton & Muellbauer (1980). Utilizing their framework and analyzing data from the Food and Agricultural Organization of the United Nations, I find that Spanish and Italian wines tend to be viewed as luxuries, while French wine is a necessity and staple in Portugal. I find French and Italian wines to be price inelastic, while Spanish wines are more elastic. Furthermore, consumers appear to view Spanish wine as highly substitutable.

2 Literature Review

The topic of alcohol demand and consumption has been widely studied in the field of economics. Some, like Gil & Molina (2009) have studied it for its policy implication. In this particular study, they use a Quadratic Almost Ideal Demand System model to investigate alcohol demand among young people in Spain. From these estimates, they suggest that a tax policy would be an efficient way to reduce consumption of alcohol among young people. Others have studied it in order to simply understand the alcohol market's mechanics. Cei & Rosetto (2024), for instance, also utilized a Quadratic Almost Ideal Demand System (QUAIDS) model to estimate demand in the sparkling wine industry in Europe. They find the sparkling wine variety to be a luxury good that is highly price sensitive. Blake & Nied (1997) study alcohol demand in the United Kingdom specifically, and explore the relationship between expenditure shares and certain demographic characteristics as well as marketing variables. Across the pond, Eakins & Gallagher (2003) use the AIDS model to estimate demand for beer in spirits in Ireland. As expected, they find both to be highly price inelastic in the Irish economy. Guoveia & Rebelo (2020) studied worldwide demand for alcoholic beverages and find beer and spirits to be complements and the pair to be imperfect substitutes of wine.

While many papers in the literature utilize the AIDS model to estimate alcohol demand, this study is unique in that it takes the approach of Aljohani, et. al. (2025) and estimates specifically the import demand. Aljohani, et. al. performed a study on demand for coffee in Saudi Arabia, utilizing the phenomenon of locational preference to study country-level data with an Almost Ideal Demand System model. Their results show strong locational preference for coffee from Ethiopia over other markets. This paper follows this model in attempting to find locational preference in wine demand (in this case, focusing specifically on Portugal's demand), something the alcohol literature is yet to explore and that has many implications for policymakers.

3 Data and Method

The data collected comes from the Food and Agricultural Organization (FAO) of the United Nations. I collected Portugal's wine import data for the years of 1987 to 2023. While I collected the data for all imports from all countries, in this study I am specifically interested in the imports from Spain, France, and Italy and, therefore, combine the imports of all other countries into a fourth aggregated good called "other." The price of this "other" good is found by quantity weighting all the importing countries' prices and finding an aggregated price. The summary of this data is given in Table 1. Figure 1 shows the historic trends of quantity of wine imported into Portugal from each country. I have used log quantity on the vertical access to scale down the quantity due to Spain's dominance in the import market.

The model I use is the Almost Ideal Demand System (AIDS) which was developed by Deaton and Muellbauer (1980). The AIDS model represents expenditure share as a function of prices and total expenditure:

$$w_i = \alpha_i + \sum_{j=1}^{n} \gamma_{ij} \ln p_j + \beta_i \ln \left(\frac{X}{P} \right)$$

Country	Quantity (tons)			Country	Expenditure (1000 USD)		
	Min	Mean	Max		Min	Mean	Max
France	78	1407	3245	France	635	13469	39084
Italy	6	8893	51581	Italy	21	11881	36031
Spain	50	130081	277680	Spain	144	65635	143578
Other	58	2052	26509	Other	160	4312	10610
Country	**Price (1000 USD/ton)**			**Country**	**Budget Share**		
	Min	Mean	Max		Min	Mean	Max
France	1.73	9.42	19.93	France	0.015	0.176	0.711
Italy	0.41	2.12	4.09	Italy	0.014	0.158	0.512
Spain	0.31	0.71	3.68	Spain	0.056	0.610	0.773
Other	0.39	3.62	8.41	Other	0.009	0.056	0.260

Table 1:

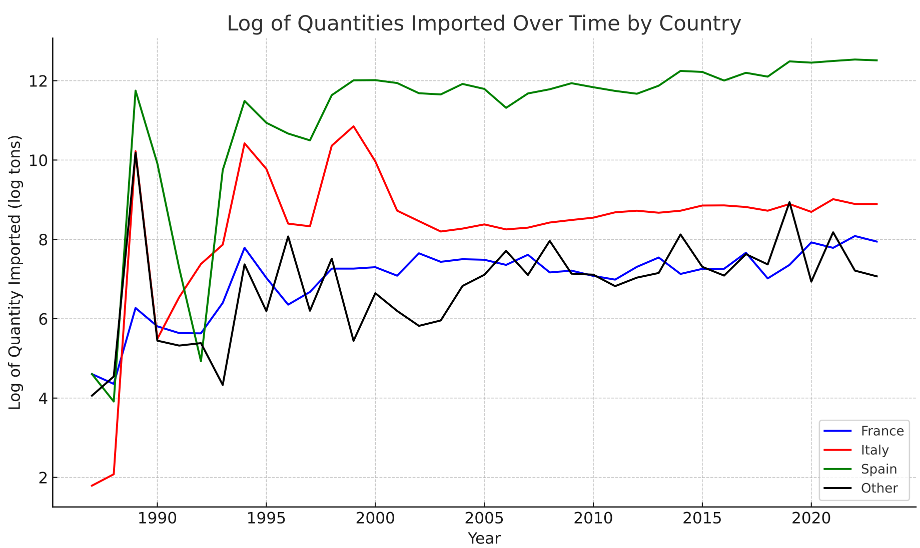

Figure 1:

where w_i is the expenditure share for each good and p_i is the price of each good, capital X is the total expenditure, and capital P is equal to the price index. When doing a Linear Approximation AIDS (LA-AIDS) model, this price index can be estimated in several ways, but most popular in empirical work because of its ease and simplicity is the Stoneman Index. I, however, am using the full AIDS model and the translog price index:

$$\log P = \alpha_0 + \sum_{k=1}^{n} \alpha_k \log p_k + \frac{1}{2} \sum_{j=1}^{n} \sum_{k=1}^{n} \gamma_{jk} \log p_j \log p_k$$

Based on microeconomic theory I also restrict my model to meet the condition of both homogeneity and symmetry:

$$\textstyle\sum_{j=1}^{n} \gamma_{ij} = 0 \;\forall i \quad \text{(Homogeneity)}$$

$$\gamma_{ij} = \gamma_{ji} \;\forall i, j \quad \text{(Symmetry)}$$

From the parameter estimates of this model $(\alpha_i, \beta_i, \gamma_{ij})$ I can use the equations from Green and Alston (1990) to derive the demand elasticities in my model:

Expenditure Elasticity

$$\eta_i = 1 + \frac{\beta_i}{w_i} \tag{1}$$

Hicksian (Compensated) Price Elasticity

$$\epsilon_{ij}^{h} = \frac{1}{w_i} \left(\gamma_{ij} - \beta_i \cdot (\alpha_j + \sum_{k} \gamma_{jk} \log p_k) \right) \tag{2}$$

Marshallian (Uncompensated) Price Elasticity

$$\epsilon_{ij} = \delta_{ij} - 1 + \frac{1}{w_i} \left(\gamma_{ij} - \beta_i \cdot (\alpha_j + \sum_{k} \gamma_{jk} \log p_k) \right) \tag{3}$$

4 Results

Using the micEconAids package in R from Henningsen & Hamann (2008) I utilize the Almost Ideal Demand System to estimate the import demand for wine in Portugal. The results are shown in Table 2 and 3. From these tables we can make several conclusions. The first and most important is to recognize that wine is clearly not considered a homogeneous product. This is clear from the widely varying expenditure and price elasticities. Italy and Spain both exhibit expenditure elasticities greater than one ($\eta_i > 1$), implying they are luxury goods. France, on the other hand, exhibits an expenditure elasticity close to 0.50, implying it is a strong necessity. The results also adhere to the Law of Demand (negative own-price elasticities) with France and Italy appearing to be relatively price inelastic - meaning demand for French wine is not very responsive to price. Spain, however, appears price elastic - meaning demand is strongly sensitive to price.

	ϵ_{i1}	ϵ_{i2}	ϵ_{i3}	ϵ_{i4}	η_i
France	-0.578^{**}	-0.171	$0.169^{\cdot}$	0.099	0.481^{***}
Italy	-0.287	-0.875^{*}	0.200	-0.058	1.020^{***}
Spain	-0.074^{*}	0.025	-1.127^{***}	-0.007	1.183^{***}
Other	0.293	-0.092	0.286	-1.066^{***}	0.579^{***}

Table 2: Marshallian (Uncompensated) Elasticities

	ϵ^h_{i1}	ϵ^h_{i2}	ϵ^h_{i3}	ϵ^h_{i4}
France	-0.493^{*}	-0.095	0.463^{***}	0.126
Italy	-0.107	-0.714^{*}	0.822^{***}	-0.001
Spain	0.134^{***}	0.213^{***}	-0.405^{***}	$0.059^{\cdot}$
Other	0.395	-0.001	$0.639^{\cdot}$	-1.033^{***}

Table 3: Hicksian (Compensated) Elasticities

The Hicksian elasticities also show that Spanish wine appears highly substitutable. This implies that consumers allocate their demand to and away from Spanish wine readily based on the price ratio with other countries. Based on this high substitutability and an elastic own-price elasticity, it makes sense that from the data we see Spain has the lowest priced wine.

5 Conclusion

This study employs the Almost Ideal Demand System to estimate Portugal's demand for wine imported from the world's three largest producers - Spain, France, and Italy. The results show significant and negative own-price elasticities, with French wine being a necessity and Spanish and Italian wines being luxuries. The results also show Spanish wine to be highly substitutable with other countries.

Wine production in Spain, France, and Italy is of the highest importance to those economies. Policymakers would do well to understand the intricacies of import demand for these countries' wines. This paper analyzes these demand relationships and highlights the many intricacies of the import market. Policymakers would benefit from knowing that French wine is considered a staple good in Portugal. I suspect they would also be interested in better understanding the substitutability of wines from these three countries, and as a consequence, the market power enjoyed by each. Finally, all three of these exporting countries can use these results to help predict the demand for their wine in Portugal in the years to come. For instance, if incomes increase in Portugal this would greatly increase demand for Spanish and Italian wines, which may play an important role in predicting import demand from these countries. These results provide empirical insights that can be applied broadly for policies on taxation and tariffs, as well as pricing strategies and marketing by exporters.

6 References

Aljohani, E. S., Chidmi, B., Kotb, A., Alderiny, M., Aldakhil, A., and Krimly, Y. (2025). Estimating the demand for imported green coffee in Saudi Arabia using the Almost Ideal Demand System. *Frontiers in Sustainable Food Systems*, 9:1516742. `https://doi.org/10.3389/fsufs.2025.1516742`

Blake, D., and Nied, A. (1997). The demand for alcohol in the United Kingdom. *Applied Economics*, 29(12), 1655–1672. `https://doi.org/10.1080/00036849700000041`

Cei, L., and Rossetto, L. (2024). The demand for sparkling wine: insights on a diversified European market. *International Journal of Wine Business Research*, 36(4), 505–527. `https://doi.org/10.1108/IJWBR-07-2023-0038`

Deaton, A., and Muellbauer, J. (1980). An almost ideal demand system. *American Economic Review*, 70, 312–326.

Eakins, J. M., and Gallagher, L. A. (2003). Dynamic almost ideal demand systems: an empirical analysis of alcohol expenditure in Ireland. *Applied Economics*, 35(9), 1025–1036. `https://doi.org/10.1080/0003684032000066796`

Gil, A. I., and Molina, J. A. (2009). Alcohol demand among young people in Spain: an addictive QUAIDS. *Empirical Economics*, 36, 515–530. `https://doi.org/10.1007/s00181-008-0209-y`

Green, R., and Alston, J. M. (1990). Elasticities in AIDS Models. *American Journal of Agricultural Economics*, 72(2), 442–445.

Henningsen, A., and Hamann, J. D. (2008). Systemfit: A package for estimating systems of simultaneous equations in R. *Journal of Statistical Software*, 23, 1–40. `https://doi.org/10.18637/jss.v023.i04`